Mapping of the Logical Structure of Partial Differential Equations using an Agentic Knowledge Graph Framework

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Motivation

Partial Differential Equations (PDEs) are foundational to modern science and engineering. However, the literature in this field is vast and deeply interconnected, making it a significant challenge to discern the complex relationships between equations, theorems, and numerical methods. This reliance on manual exploration and expert intuition slows down progress and raises the barrier of entry for newcomers. To address this knowledge navigation problem, this thesis proposes a computational approach to systematically map the logical structure of PDE research.

Proposed Framework

This thesis introduces a novel agentic Knowledge Graph (KG) framework tailored to the unique demands of mathematical literature. The approach builds upon recent advances in multi-agent AI for scientific discovery, employing a pipeline of specialised, Large Language Model (LLM)-powered agents to parse a curated corpus of PDE research papers. The agents' core tasks are to:

- Extract key entities (e.g., named equations, theorems, solution techniques).
- Identify the logical and semantic relationships between them.
- Organise this information into a structured, machine-readable Knowledge Graph.

The research will concentrate on the domain of **hyperbolic wave problems** and their associated advanced numerical techniques, including **Finite Element Methods (FEM), Discontinuous Galerkin (DG), Hybridised DG (HDG), and High-Order Hybrid (HHO) methods**.

Anticipated Contributions

The central contribution will be the **design and implementation of the agentic system** and the resulting **Knowledge Graph** that captures the logical structure of hyperbolic PDE methods. More broadly, the thesis aims to:

- Deliver an **automated framework** for extracting and structuring mathematical knowledge.
- Produce a **domain-specific KG** interlinking PDE equations, theorems, and computational methods.
- Demonstrate how structured knowledge supports **navigation and reasoning** in advanced mathematical domains.
- Optionally, if time permits, develop a **prototype frontend tool** to showcase the practical utility of the KG.

References

- Bo, L., et al. (2024). SciAgents: A Multi-Agent System with a Knowledge Graph for Scientific Discovery. arXiv preprint.
- Hogan, A., et al. (2021). Knowledge graphs. ACM Computing Surveys (CSUR), 54(4), 1-37.